Book Data API

COMP3011 - Web Services and Web Data

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https://github.com/Daiaye/book-api

Technical Foundation

Justifying the technology stack for robustness and performance.

Language & Framework

Python

Concise syntax

Robust data science ecosystem future-proofs for future machine learning recommendation engines.

FastAPI

High-performance framework utilising Python Type Hints for automatic validation

Industry standard and automatic documentation generation.

SQLite Database

Zero Configuration and Network Latency

Serverless nature makes it extremely portable

Utilises a file based storage system

Acid Compliance

Guarantees structural integrity during concurrent CRUD operations

Architectural Design

Focusing on performance optimisation and security resilience.

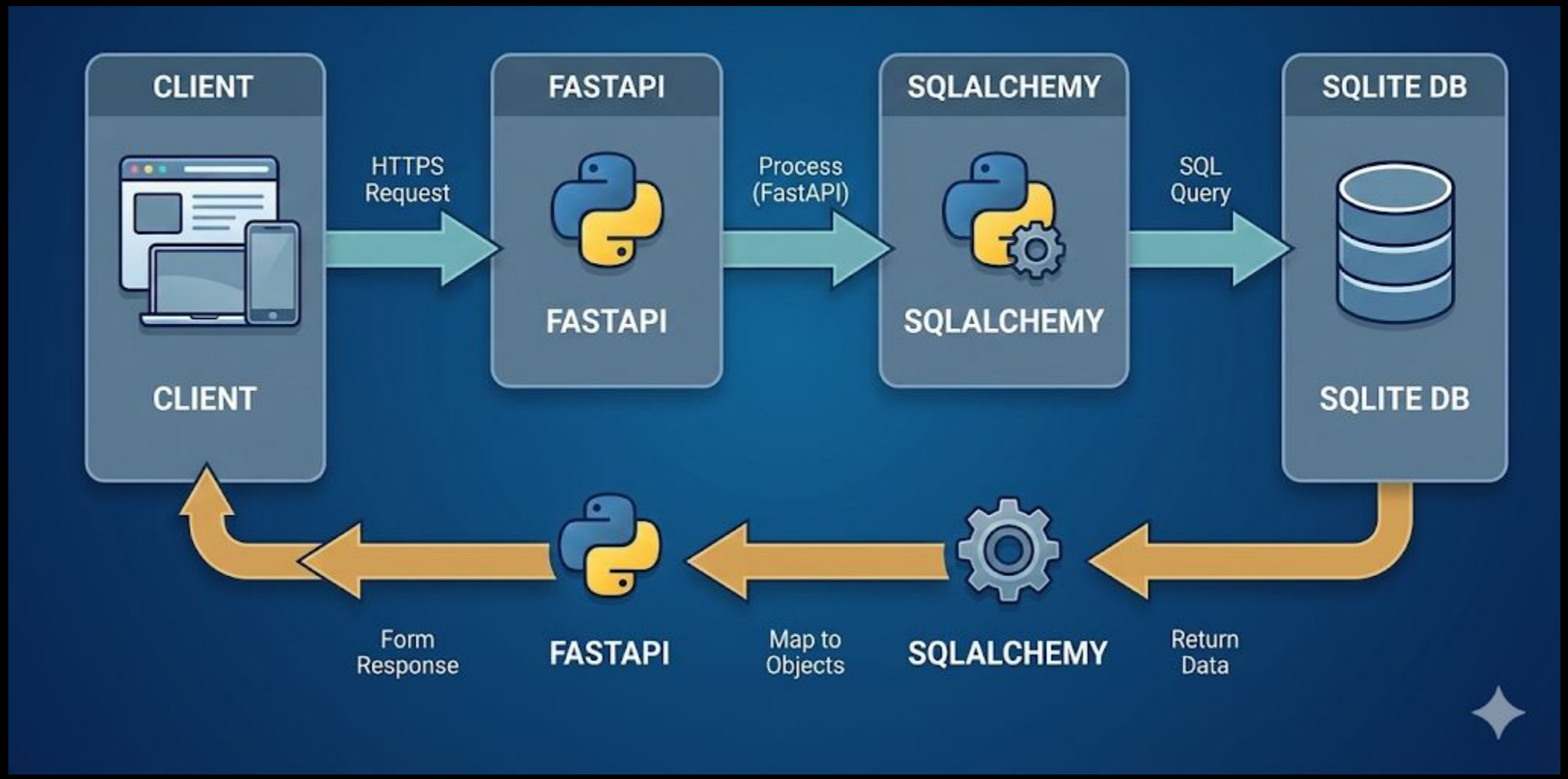


Image Credit: Gemini Nano Banana

Optimisation & Security

```python
# Blueprint for updating an existing book
class BookUpdate(BaseModel):
    title: str = None
    author: str = None
    year: int = None
    publisher: str = None
```

Denormalisation

Pre-calculated 'average_rating' during data ingestion

Ensures fast read times for static metadata

Resilient Filtering

Implemented schema-based filtering to prevent Mass Assignment vulnerabilities

Silently discards unauthorised JSON parameters

```python
# Update the fields if they were provided
if book_update.title: db_book.title = book_update.title
if book_update.author: db_book.author = book_update.author
if book_update.year: db_book.year = book_update.year
if book_update.publisher: db_book.publisher = book_update.publisher
```

Version Control Strategy

Atomic Commits: Maintained a descriptive commit history

Dependency Management: requirements.txt for deterministic environment reproducibility

Environment Isolation: Use of .venv to prevent library conflicts

requirements.txt

```
annotated-doc==0.0.4
annotated-types==0.7.0
anyio==4.12.1
click==8.3.1
colorama==0.4.6
fastapi==0.135.1
greenlet==3.3.2
h11==0.16.0
idna==3.11
numpy==2.4.2
pandas==3.0.1
pydantic==2.12.5
pydantic_core==2.41.5
python-dateutil==2.9.0.post0
six==1.17.0
SQLAlchemy==2.0.48
starlette==0.52.1
typing-inspection==0.4.2
typing_extensions==4.15.0
tzdata==2025.3
uvicorn==0.41.0
```

.venv
- Include
- Lib
- Scripts
- pyvenv.cfg

Commits on Mar 2, 2026

Create database
Daiaye committed last week

Add .gitignore file
Daiaye committed last week

Initial commit
Daiaye authored last week

Commits on Mar 3, 2026

Create GET /books endpoint for a single book
Daiaye committed last week

Create PUT /books endpoint
Daiaye committed last week

Create POST /books endpoint
Daiaye committed last week

Create GET /books endpoint
Daiaye committed last week

Update .gitignore file
Daiaye committed last week

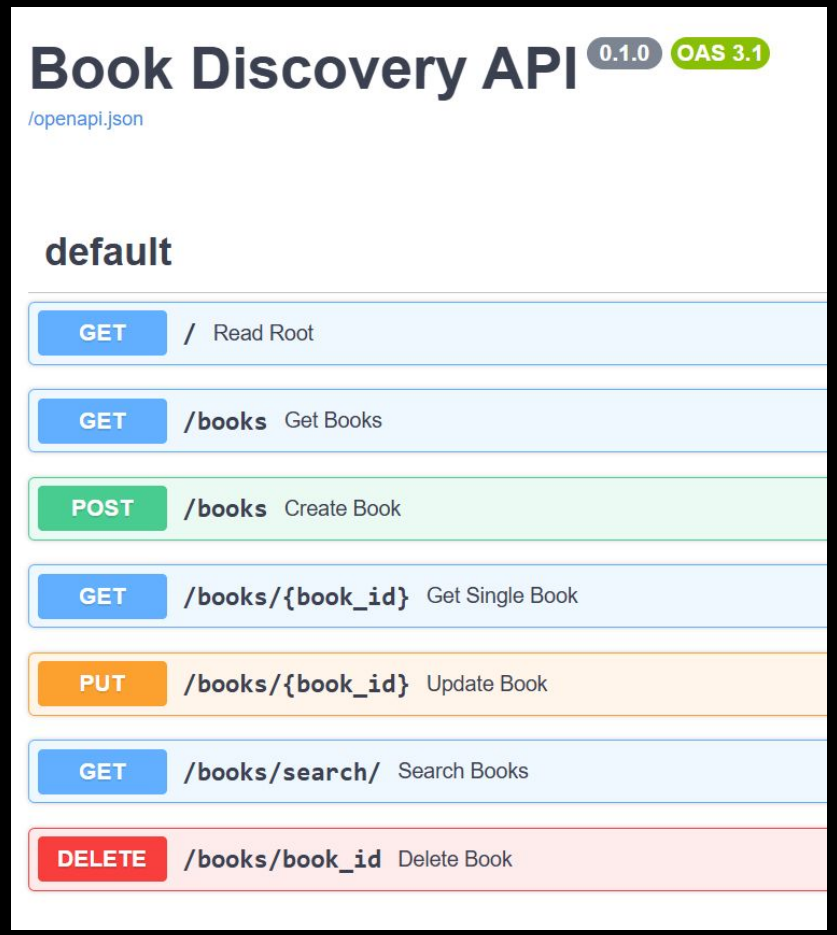
Add script to import data
Daiaye committed last week

Interactive Documentation

Live Testing: Immediate verification of endpoint logic through the automated /docs interface.

Clarity: Explicitly defined JSON request/response bodies for all CRUD operations.

Error Mapping: Standardised documentation for 404, 422, and 201 status codes.

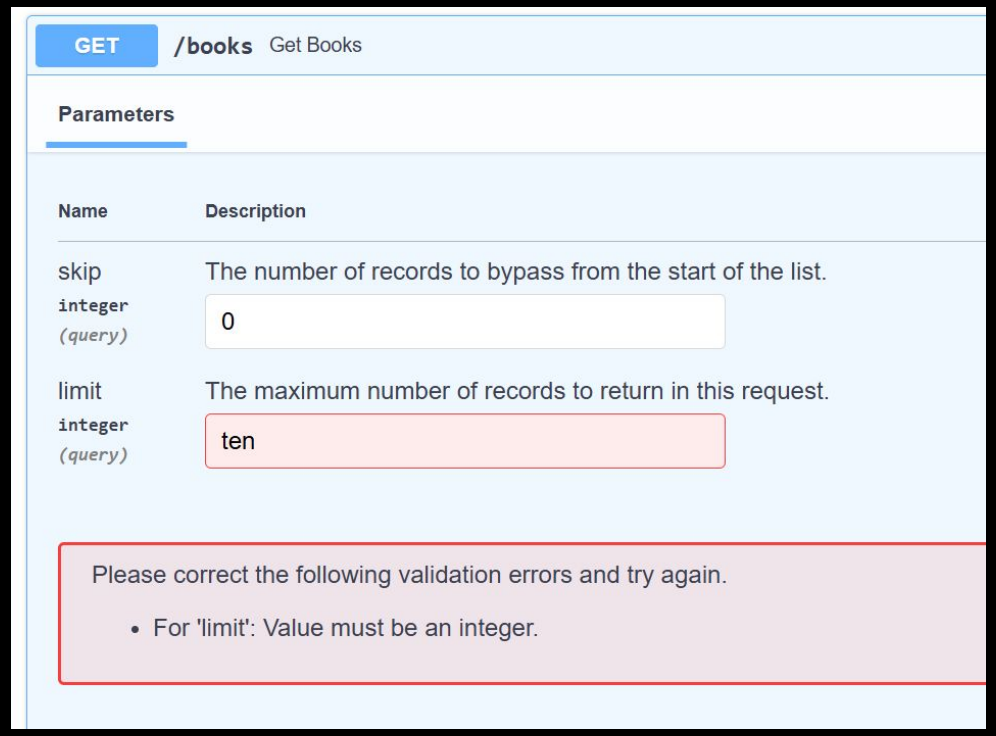


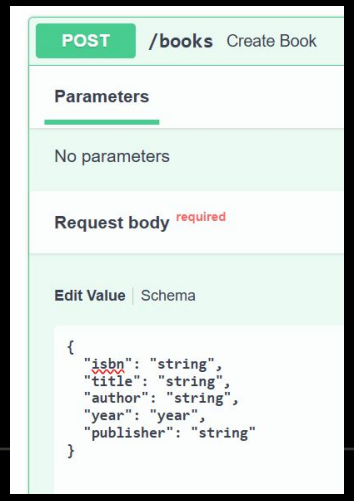
Testing

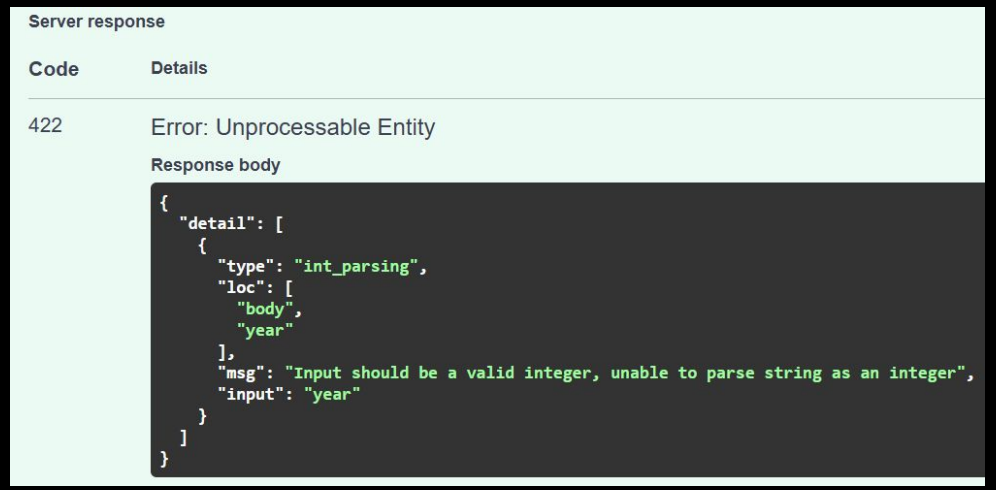
Negative Testing: Verified that Pydantic rejects invalid types with 422 errors.

Graceful recovery: Uses 404 responses when non-existent IDs are requested

Data Integrity: Use of schemas to ensure valid updates only







Demo